DESIGN BRIEF

DATE: 23/10/2026

PREMISE

A shoot ’em up where every shot you fire stays in orbit.

RESEARCH

Space Is Getting Crowded

Earth’s orbit is becoming a junkyard. Dead satellites, discarded rocket stages, and fragments from collisions circle the planet at thousands of kilometres per hour. Even a tiny fragment can destroy a spacecraft at those speeds.

The numbers are staggering:

- 47,000+ objects are regularly tracked in orbit.
- 1.2 million fragments between 1 and 10 cm are estimated to be circling Earth.
- 140 million smaller fragments between 1 mm and 1 cm may be out there, too.

Source: European Space Agency (ESA), 2026 Space Environment Report.

The Race to Fill the Sky

Companies such as SpaceX (Starlink), Amazon, and Eutelsat OneWeb are racing to fill orbit with satellites. Their networks promise global internet and communications, but they also serve a commercial race for market share, infrastructure, and long-term control of valuable services.

The costs are not shared equally. Companies can profit from deploying more satellites, while the risks of collisions, orbital congestion, and damage to astronomy affect a much wider community.

Consider the scale:

- 1.7 million proposed satellites: Plans submitted for future satellite deployments could dramatically change the night sky and threaten astronomical research.
- 300,000+ collision-avoidance manoeuvres: Starlink satellites reportedly performed more than 300,000 manoeuvres in 2025 alone.

Orbit is a shared environment, but decisions about what goes into it are made by a relatively small number of companies and governments. The rules are struggling to keep pace with the expansion.

Sources: European Southern Observatory (ESO), 2026; OECD, The Space Economy at a Glance 2026.

Kessler Syndrome: When Space Junk Makes More Space Junk

In 1978, scientist Donald Kessler described a troubling possibility: collisions create debris, debris causes more collisions, and the cycle continues. Some orbital regions eventually become impossibly dangerous to use. This is the "**Kessler Syndrome**".

We've already seen how things can go wrong:

- 2007 — Fengyun-1C: China destroyed one of its own weather satellites in an anti-satellite test, creating thousands of debris fragments. These fragments can still be tracked to this day.
- 2009 — Iridium 33 and Cosmos 2251: Two satellites collided by accident, creating a large cloud of debris, taking out a part of usable orbit.

Cleaning up is difficult and expensive. Researchers have explored **nets**, **harpoons**, and **robotic arms** to capture dead satellites and other large objects. But removing debris after the fact is harder than preventing it in the first place. What happens when everyone wants to get ahead, but nobody can afford to ignore the mess?

The Space Race

The competition between the USA and USSR turned space into a stage for technological achievement, political power, and national pride.

- 1957 — Sputnik 1: The Soviet Union launches the first artificial satellite.
- 1961 — Yuri Gagarin: The first human travels into space.
- 1969 — Apollo 11: The USA lands humans on the Moon.

Each achievement raised the stakes. Space was a frontier to explore, but also a place to compete, demonstrate power, and make history. The space race provides some historical lens for thinking about competition. Some may have a strong incentive to achieve an immediate objective and care little about the long-term. What happens when the race to do more leaves a mess for everyone else?

REFERENCES

Space Invaders (1978)

The invaders advance, the pressure builds, and even your protective shields can be destroyed by your own shots.
Look for: How a few simple rules create escalating tension.

Asteroids (1979)

Shoot a large asteroid and it splits into smaller, more dangerous pieces. Destroying a threat can create several new ones.
Look for: How solving one problem can create more problems.

Geometry Wars (2003)

Fast movement, immediate controls, and an arena packed with threats.
Look for: How movement and shooting can feel responsive and satisfying.

THEME

Earth's orbit is crowded. Satellites compete for space, debris drifts through the darkness, and collisions threaten to trigger a chain reaction.

You pilot a small craft tasked with keeping orbit clear. But your tools have consequences. Every shot changes the field, and every broken object leaves something behind.

Survive the mess you make.

AESTHETICS

- Challenge: The situation grows harder as debris accumulates.
- Responsibility: Your actions create future dangers.
- Escalation: A quick solution can become a bigger problem.
- Clarity: Players should understand how their actions change the situation.

This isn't a power fantasy about clearing the screen. It's about making decisions in a system that remembers them.

DESIGN QUESTION

How can shooting change the space the player plays in, making every shot a decision while keeping the game fast and readable?

Consider

- What counts as debris, and how it behaves. Is there anything else?
- Whether debris can be cleared, collected or used, and how.
- Whether "shooting" is the only verb. Should it even be a verb?
- What the player scores, and how a run ends.

Start with the central interaction. Prototype different answers before committing to one. *Remember: You don't need to simulate orbital mechanics. You need to make the consequences of shooting interesting.*



MOODBOARD











